

STEFANO BUDI

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EDUCATION

BINUS University

B.S. in Computer Science, Data Science Specialization Track

GPA: 3.97 / 4.00

Independent Study: MIT 15.401, Harvard STAT110, Stanford CS229

Jakarta, Indonesia

2024 – 2028 (Expected)

PUBLICATIONS

“Dynamic Classifier Selection using Local Class Accuracy on an Overlap-Cleaned Heterogeneous Pool of Experts”

Sept 2026

2nd Author | Published

- Co-developed a dynamic classifier selection framework combining KNN-based overlap cleaning with local-class-accuracy-based expert selection across a heterogeneous pool of specialists (Naive Bayes, XGBoost, SVM), cutting log loss by 70.1% (0.898 → 0.269) versus a probability-harvesting baseline while achieving 91.62% accuracy and a macro F1-score of 0.8638 on the UCI Student Dropout dataset (n=4,424).

EXPERIENCE

Laboratory Assistant, BINUS University

Feb 2025 – Feb 2026

- Selected as 1 of 2 assistants from ~200 applicants (top <1%) through a 4-month, 3-stage process, including 1 month of full-time (7am–5pm) on-site bootcamp training, project creation, and presentation.
- Taught 5 classes weekly (~150 students) across Computer Vision and Algorithm & Programming (C), qualified in Deep Learning and Data Mining via BINUS's SLC qualifications; earned a 5.3/6 performance KPI.

PROJECTS

Speech2Market — End-to-End NLP-Based Market Impact Predictor for FED Speeches

Aug 2026

[Dashboard](#) | [Live App](#) | [GitHub](#)

- Engineered an NLP + time-series pipeline (classifier-less FinBERT embeddings, lagged feature engineering, macroeconomic indicators, HistGradientBoost) predicting SPX, Gold, VIX, and TNX returns at 3/7/30-day horizons following The FED's speeches.
- Deployed under a 1GB memory constraint via FP16 ONNX quantization of FinBERT, and built an automated GitHub Actions pipeline to work around FRED API/Yahoo Finance API blacklist, running entirely on free-tier infrastructure.

FindIT — Face Anti-Spoofing Detection

Apr 2026

Computer vision competition — camera-based identity fraud detection (DAC Find IT! 2026)

- Built a 5-fold DINOv2-based face anti-spoofing classifier across 6 classes using a two-phase fine-tuning strategy — frozen-backbone head training followed by full fine-tuning with differential learning rates — plus weighted sampling and selective MixUp to address class imbalance.
- Achieved 0.9742 macro F1, placing 11th of 340 teams (top 4%) within 0.005 of 1st place.

Laptop Price Predictor — End-to-End Ensemble ML Deployment

May 2026

[Live App](#) | [GitHub](#)

- Built a stacking ensemble (XGBoost, LightGBM, CatBoost → RidgeCV meta-learner) tuned via Optuna Bayesian optimization, achieving 6.67% MAPE and outperforming all base models in accuracy and cross-validation stability.
- Shipped a full pipeline from training with asynchronous web-scraped dataset (BeautifulSoup + Playwright, 583 listings) to a deployed Streamlit app (P90 latency ~65–75ms), validated via a 5-user study with 100% “useful” rating.

MBG Investment Priority Mapping — Graph-Based Clustering Research Project

Jul 2025

[Dashboard](#) | [GitHub](#)

- Applied spectral (graph-based) clustering with a Laplacian structure to map investment priority across 38 Indonesian provinces, engineering 15+ derived features from raw government data (BPS, Badan Pangan Nasional), filtered via RFECV, validated via the Calinski-Harabasz index (3.94 vs. a 1.01 random-baseline benchmark); selected by BINUS's Computer Science faculty as product presentation material for a public open event.

TECHNICAL SKILLS

Languages: Python, SQL (fluent); C (proficient); Java (basic); RapidMiner, R (familiar)

Machine Learning: Gradient Boosting, Spectral Clustering, Temporal Fusion Transformers, Naïve Bayes, Decision Trees, SVM, PCA/LDA, Optuna

Transfer Learning: DINOv2, ConvNeXt, FinBERT, ONNX Runtime, LLM Application Integration

Data Engineering & Deployment: Pandas, BeautifulSoup, Playwright, GitHub Actions, Streamlit, Git/GitHub